

Low-Quality Training Data Only?

A Robust Framework for Detecting Encrypted Malicious Network Traffic

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Tannu Vijay
(2023UCP1620)

Chainy Jain
(2023UCP1620)



3 real-world problems in malware detection

- Bad training data (wrong labels) → solved by label correction
- Too little data → solved by data augmentation (GAN)
- Hard-to-detect encrypted traffic → solved by smart feature extraction



ML/DL-based Encrypted Malicious Traffic Detection

- ML/DL enables detection of encrypted malicious traffic
- Traditional payload-based methods fail with encryption
- Models learn traffic patterns instead of content
- Heavy reliance on high-quality labeled data
- Data collection and labeling remain challenging

What if we only have **Low-quality** training data?



What's Low-Quality Training Data

Label Noise Problem

- Inconsistent labels (e.g., VirusTotal)
- Changes over time
- Manual labeling is expensive & difficult

Limited Malicious Data

- Collected via sandbox execution
- Not fully realistic
- Hard to keep up with evolving malware

Low-quality training data can degrade the detection performance of ML/DL-based encrypted malicious traffic detection methods.



Potential Solution?

- Robust Machine Learning Models
 - Rely on strong assumptions or prior knowledge.
- Data Augmentation Methods
 - Label noises will confuse the distributions of different categories, resulting in synthesizing more label noises
- Pre-training an DL model using large-scale unlabeled encrypted training data
 - Collecting and pre-processing a large-scale dataset is expensive. It may also increase the risk of privacy leakage.

- Our Goal: **Detecting encrypted malicious traffic accurately using only a low-quality training set.**



Feature Extraction Module

Goal: Convert encrypted network traffic into meaningful feature representations

Challenges:

- Works with different encryption protocols & versions
- Label noise can affect feature quality

Input Representation :

- Packet length sequence of each traffic flow
- Captures sequential traffic behavior



Label Correction Module

Goal: Estimating training data's distributions to detect and correct label noises

Challenge: High-dimensional traffic data's distribution is hard to accurately estimate & The difference between normal and malicious data distributions may not be significant

Using a **Deep Generative Model** to accurately estimate the data's distribution, relabeling part of the training samples that have the most significant distributions, and utilizing them to infer other samples' labels through ensemble learning

- Uses MADE (Masked Autoencoder for Distribution Estimation)
- Learns the underlying distribution of traffic data

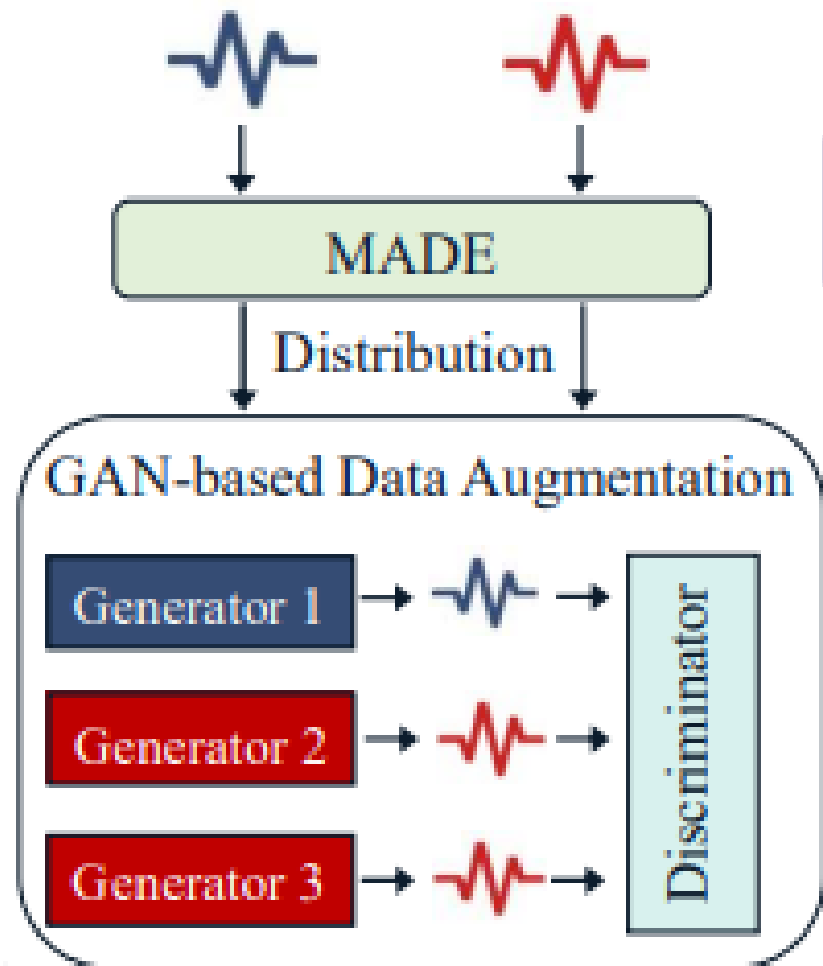


Data Augmentation Module

Goal: Generate new training data that can represent new encrypted malicious traffic

Challenge The generated data should be diverse, otherwise it may limit the model's generalizability & New malicious samples' distribution may be inconsistent with existing training data

Data Augmentation Module



Predicting the potential regions of new malicious samples,
sampling from these target regions to generate new diverse training data



Simulating Real-World Label Noise

Why simulate label noise?

To reflect **real-world inconsistencies** in labeling and evaluate model robustness

Benign → Misclassified

- Normal traffic flagged as malicious
- Caused by unusual patterns or unclear behavior

Malicious → Misclassified

- Malicious traffic seen as benign
- Happens due to missing or outdated threat information



We used CICIDS DATASET in pcap format and by using this the

Recall: 0.75999999999848

Precision: 0.7900207900043654

F1: 0.7747196687883238

Accuracy: 0.9598181818164367



Novel Work - Co-teaching

Co-Teaching Technique

- Two MLP neural network models are trained simultaneously.
- Each model selects low-loss (reliable) samples during training.
- The selected clean samples are exchanged between both models.
- Noisy or incorrect samples are gradually ignored using a forget-rate strategy.
- This improves robustness, reduces overfitting, and enhances generalization.
- Co-teaching helps improve intrusion detection performance on noisy and imbalanced datasets.



Novel Work - Co-teaching

Advantages

- Better handling of noisy labels
- Improved attack detection capability
- Higher recall and model stability
- More generalized learning compared to single-model training



This is the output after novel work

Recall: 0.93999999999812

Precision: 0.6216931216848982

F1: 0.7484076385079009

Accuracy: 0.9425454545437408



Thank You

